

Extracted Content

TABLE I
nFAP Delay Management Timing Offset Parameters [1]

Parameter	Description
Timing offsets (DL_TTI, UL_TTI, UL_DCI, Tx_Data)	The timing offset (μs) before the slot start that the corresponding request must be received at PHY.

TABLE II
nFAP Timing Info Parameters [1]

Field	Description
Last SFN / Slot	Triggering SFN/Slot.
Time Since Last Info	Time since last report (ms).
Jitter	DL_TTI, TX_DATA, UL_TTI, UL_DCI.
Latest Delay	DL_TTI, TX_DATA, UL_TTI, UL_DCI from the latest time acceptable (μs).
Earliest Arrival	DL_TTI, TX_DATA, UL_TTI, UL_DCI from the latest time acceptable (μs).
Subcarrier Spacing	SCS index (TS38.211).

TABLE III
nFAP Delay Management Configuration Parameters

Parameter	Description
timingWindow	Duration of the sliding window used for delay calculation.
timingMode	Reporting mode for Timing Info: Periodic, Aperiodic, or Both.
timingPeriod	Transmission interval for Timing Info messages.

$$\Delta t_{offset} = \frac{(t_2 - t_1) - (t_4 - t_3)}{2} \quad (1)$$

$$T_{owd} = \frac{(t_4 - t_1) - (t_3 - t_2)}{2} \quad (2)$$

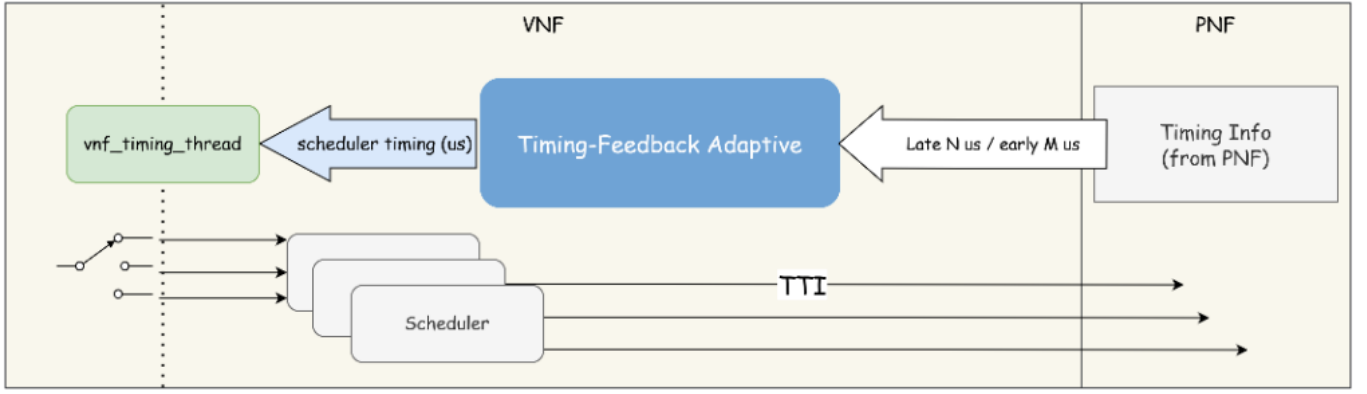


Fig. 1. System Model of the Timing-Feedback Adaptive Scheduler.

$$\Delta t_{total} = \Delta t_{offset} + T_{margin} \quad (3)$$

$$N_{slot} = \lfloor \frac{\Delta t_{total}}{T_{slot}} \rfloor \quad (4)$$

$$T_{us} = \Delta t_{total} \pmod{T_{slot}} \quad (5)$$

$$A_{us} = -T_{us} \quad (6)$$

$$A_{slot} = N_{slot} \quad (7)$$

$$N_{skip} = \lfloor \frac{\Delta t_{accumulated}}{T_{slot}} \rfloor \quad (8)$$

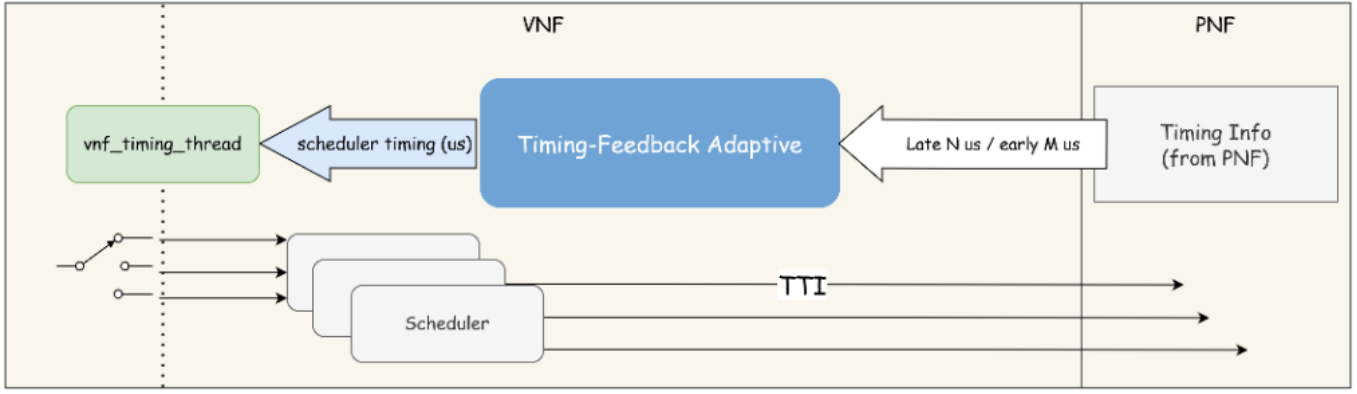


Fig. 2. Receiving Window mechanism at the PHY

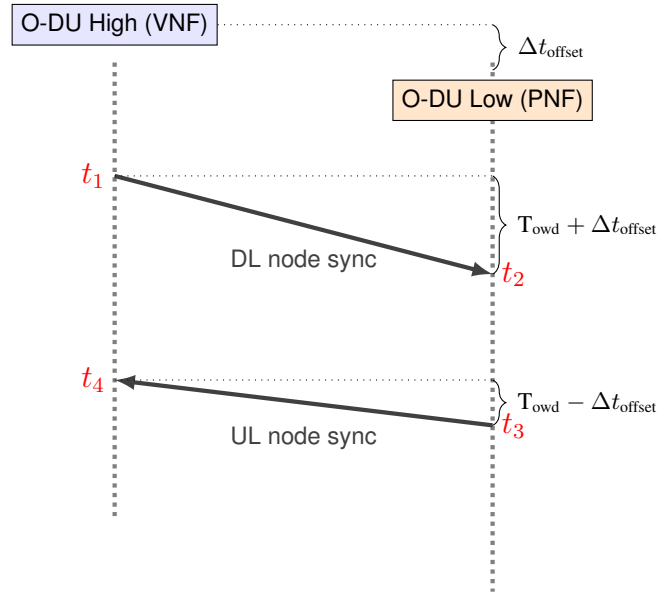


Fig. 3. Delay Management Procedure (Node Sync)

TABLE IV
DL Node Sync parameters

Field	Description
t1	Offset from VNF SFN/slot 0/0 (0..10239999 μ s).
Delta SFN/slot	Slot shift update (-163839..163839). Negative pulls timeline.
Sub-carrier spacing	SCS index (0: 15kHz .. 4: 240kHz).

TABLE V
UL Node Sync parameters

Field	Description
t1	The supplied t1 field from DL Node Sync (0..10239999 μ s).
t2	Reception time of DL Node Sync at PNF (0..10239999 μ s).
t3	Transmission time of UL Node Sync at PNF (0..10239999 μ s).

References

- [1] Small Cell Forum, “5g nfapi specification,” Small Cell Forum, Tech. Rep., July 2022, table 4-3: Timing Info parameters.